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Received: 7 November 2025

Accepted: 26 March 2026

Published online: 03 April 2026

Cite this article as: Jaiswal M., Nagwanshi K.K., Khan M.H. *et al.* YOLOv8s-WAMNet: enhancing robust vehicle detection under adverse weather via hybrid attention and multi-scale fusion in real time. *Sci Rep* (2026). <https://doi.org/10.1038/s41598-026-46527-5>

Manjit Jaiswal, Kapil Kumar Nagwanshi, Maleika Heenaye-Mamode Khan, Upendra Verma & Amelia Taylor

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YOLOv8s-WAMNet: Enhancing Robust Vehicle Detection under Adverse Weather via Hybrid Attention and Multi-Scale Fusion in Real Time

Manjit Jaiswal¹, Kapil Kumar Nagwanshi^{1*}, Maleika Heenaye-Mamode Khan²,
Upendra Verma³, Amelia Taylor^{4*}

¹Department of Computer Science and Engineering, Guru Ghasidas Vishwavidyalaya, Bilaspur, India.

²Faculty of Information, Communication and Digital Technology, University of Mauritius, Reduit, Mauritius.

³Department of Computer Science, NMIMS University, Mumbai, India.

⁴Department of Computing and IT, Malawi University of Business and Applied Science, Blantyre, Malawi.

*Corresponding author(s). E-mail(s): dr.kapil@ieee.org; ataylor@mubas.ac.mu;
Contributing authors: manjit.jaiswal222@gmail.com; m.mamodekhan@uom.ac.mu;
upendra.nmims@gmail.com;

Abstract

Vehicle detection in adverse weather is crucial for autonomous driving; however, fog, rain, snow, and low illumination significantly degrade feature quality and detection reliability. This paper presents YOLOv8s-WAMNet (Weather-Adaptive Multi-Scale Network), a lightweight hybrid attention framework designed to maintain robustness under visibility degradation. The model employs an Efficient Hybrid Vision Transformer (EHViT) backbone that combines convolutional and transformer-based feature extraction for resilient representation learning. Cross-Dimensional Multi-Scale Attention (CDMSA) and a Contextual Multi-Attention Fusion (CMAF) neck enhance multi-scale feature refinement and stabilize spatial-contextual reasoning in adverse scenes. A Multi-Head Dynamic Attention (MHDA) detection head with a hybrid SIoU-MPDIoU loss further improves localization accuracy and convergence stability. Extensive experiments on the WEDGE (WEather images by DALL-E GENERation) dataset demonstrate that YOLOv8s-WAMNet achieves 51.90% mAP@50 and 24.96% mAP@50–95, outperforming YOLOv8s by +6.3 mAP@50 while reducing computational cost by approximately 54% (13.24 vs. 28.7 GFLOPs). Additional evaluations on the real-world DAWN (Detection in Adverse Weather Nature) dataset confirm the robustness and cross-dataset generalization capability of the proposed model.

Keywords: adverse weather, anchor-free detection, hybrid localization loss, vehicle detection, vision transformer, YOLOv8s

1 Introduction

Object detection has been a cornerstone of computer vision, evolving significantly since the introduction of

pioneering methods like YOLO (You Only Look Once) [1]. The original YOLO framework revolutionized object detection by enabling real-time performance while maintaining high accuracy, setting the stage for

subsequent advancements. Earlier methods like Fast R-CNN [2] and Faster R-CNN [3] focused on improving accuracy and speed by introducing region proposal networks and optimizing computational efficiency. Similarly, the SSD (Single Shot MultiBox Detector) [4] provided an alternative approach by eliminating region proposals, enabling faster detection.

The evolution of YOLO has been marked by continuous improvements. YOLO9000 [5] extended the framework to support hierarchical classification and detect over 9,000 object categories. Techniques like Focal Loss [6] addressed challenges in dense object detection, particularly for imbalanced datasets. YOLOv3 [7] introduced incremental enhancements in network design, while YOLOv4 [8] optimized speed and accuracy through innovations such as Cross-Stage Partial connections and Mish activation. The development of YOLOv5 [9] and its subsequent versions, including YOLOv6 [10] and YOLOv7 [11], further pushed the boundaries of real-time detection capabilities by incorporating modular designs and advanced optimization techniques.

Recent iterations, such as YOLOv8 [12], YOLOv9, YOLOv10, and YOLOv11 [13], emphasize scalability and adaptability, targeting applications like aerial vehicle detection and weed species identification. These models integrate novel loss functions like SIOU [14] and attention mechanisms like CBAM (Convolutional Block Attention Module) [15] to enhance detection precision.

1.1 Optimizing YOLO for Environment Challenges

Recent advancements in real-time object detection have demonstrated the potential to surpass traditional YOLO-based models, as shown by Zhao et al. [16], who highlighted the effectiveness of Detection Transformers (DETRs) in challenging scenarios. However, despite their promising performance, the high computational complexity and resource demands of DETR-based models limit their applicability in lightweight and real-time environments.

Adverse weather conditions such as rain and fog significantly degrade object detection performance, particularly in outdoor and vehicular settings. To mitigate these effects, Wang et al. [17] proposed a multi-decoding deraining network that improves training robustness through quasi-sparsity. Similarly, Quan et al. [18] addressed the simultaneous removal of raindrops and rain streaks to enhance visual clarity for machine

vision tasks. Ni et al. [19] further introduced a unified framework that enables seamless transitions between rain removal and rendering, improving adaptability across varying weather conditions.

For combined rain and fog scenarios, Liu et al. [20] presented DAGL-Faster, which integrates domain-adaptive techniques for vehicle detection under adverse weather. Their work emphasizes the importance of combining domain adaptation with advanced detection frameworks. Complementary to this, He and Liu [21] proposed an optimized feature fusion strategy that enhances obstacle detection in foggy environments, aligning with the analysis by Liu et al. [22], who examined performance degradation caused by atmospheric interference.

YOLO-based models remain central to addressing environmental challenges due to their efficiency and real-time capability. Liu and Lin [23] introduced YOLO-GW for pedestrian detection in foggy traffic scenarios, emphasizing computational efficiency and accuracy. Meng et al. [24] further proposed YOLOv5s-Fog, a model specifically designed for foggy weather object detection, highlighting the importance of adapting feature extraction and attention mechanisms to environmental conditions. Lightweight detection models also continue to attract attention; for example, Xu et al. [25] explored attention mechanisms within YOLOv5 to achieve an effective balance between speed and accuracy.

Despite these advances, robust and efficient vehicle detection under adverse weather remains challenging. Conditions such as rain, fog, snow, and low illumination distort visual inputs, reduce contrast, and obscure critical features, leading to degraded detection performance. These challenges manifest in several key aspects: distorted feature acquisition due to environmental degradation, occlusions and low contrast that hinder reliable feature recognition, difficulties in detecting small or distant vehicles due to scale variation, and challenges in precise classification and localization caused by class imbalance and blurred object boundaries.

Existing YOLOv8 architectures struggle to maintain stability under severe visibility degradation because convolution-based backbones primarily capture local textures and lack long-range contextual modeling. While transformer-based variants improve global reasoning, they introduce substantial computational overhead and may lose fine-grained spatial details. Attention mechanisms such as SE and CBAM

focus on either channel-wise or spatial features independently, without jointly addressing cross-scale degradation, contextual blur, and weather-specific distortions. Consequently, existing YOLO models often exhibit feature collapse, suppression of small objects, and inconsistent bounding box localization under adverse conditions.

1.2 Contributions

The key contributions of this work are summarized as follows:

1. **Weather-Adaptive Hybrid Backbone:** A unified backbone combining EHViT with CDMSA to enhance feature representation and resilience against weather-induced feature loss.
2. **Contextual Multi-Attention Fusion (CMAF):** A fusion mechanism that reinforces global–local contextual consistency for improved small-object and long-distance vehicle detection.
3. **Multi-Head Dynamic Attention (MHDA) Head:** A dynamic decoding head that improves localization stability and detection robustness across variable weather conditions.
4. **Hybrid SIoU–MPDIoU Loss:** A loss formulation that strengthens geometric alignment and bounding box precision by combining angular, scale, and boundary consistency constraints.
5. **Performance Superiority:** Achieves higher accuracy and efficiency than recent YOLO variants (YOLOv9–YOLOv12) and RT-DETR on the WEDGE dataset while remaining lightweight enough for real-time deployment. Consistent performance gains are also observed on the real-world DAWN dataset, demonstrating strong robustness under naturally occurring adverse weather conditions.

1.3 Structure of the Paper

The remainder of the paper is structured as follows: In Section 2, we review related work, exploring existing approaches for object detection in adverse weather and highlighting gaps in current methodologies. Section 3 describes the proposed YOLOv8s-WAMNet model, detailing its architecture and innovations, including the hybrid backbone, adaptive neck, and dynamic detection head. Section 4 experimental setup and results, including WEDGE and DAWN datasets, evaluation metrics, and implementation details used to validate

the model, Ablation Study Analysis, Performance Comparison and Performance Evaluation. Finally, Section 5 concludes the paper, summarizing the future research directions and limitations.

2 Literature Review

2.1 Object Detection in Adverse Weather

A comprehensive dataset for autonomous driving under adverse weather conditions, known as WEDGE, was proposed by Marathe et al. [26]. This dataset is built using generative vision-language models, enabling multi-weather scenarios for training and evaluating object detection systems. The dataset addresses the challenges of visual degradation caused by adverse weather and serves as a benchmark for testing model robustness. Wang et al. [27] introduced a joint image restoration framework tailored for snowy weather. The approach integrates image restoration with object detection, aiming to mitigate visual distortions caused by snow and enhance detection accuracy. This work highlights the importance of preprocessing in adverse weather scenarios. Expanding on detection under snow-fall, Wang et al. [28] presented a progressive restoration and multi-feature fusion technique to improve detection accuracy. Their method effectively addresses snow occlusions and degraded visibility by leveraging a robust multi-feature fusion framework. Further extending the research on adverse weather detection, Feng et al. [29] proposed ADWNet, an innovative and enhanced detection framework based on YOLOv8. This method is specifically designed to address the challenges posed by harsh weather conditions, such as heavy rain, dense fog, and low-light environments, which are known to significantly degrade the performance of traditional object detection models. By incorporating advanced feature extraction techniques and optimized loss functions, ADWNet achieves superior robustness and efficiency in detecting objects under these adverse conditions. The framework demonstrates exceptional applicability in autonomous driving systems, ensuring reliable performance and improved safety in complex, real-world scenarios. This contribution represents a significant step forward in advancing object detection technology for challenging weather environments.

2.2 Lightweight Object Detection Models

Wang et al. [30] proposed TRC-YOLO, a lightweight object detection model optimized for mobile devices. By incorporating CBAM, dilated convolutions, and receptive field blocks, the model achieves efficient and accurate detection for small targets. This approach emphasizes the trade-off between computational efficiency and detection accuracy. In the domain of small traffic sign detection, Du et al. [31] developed TSD-YOLO, an enhanced version of YOLOv8. The model introduces architectural improvements to address challenges specific to small object detection, particularly in traffic scenarios. The work demonstrates the adaptability of YOLO-based architectures to domain-specific tasks.

Huang et al. [32] proposed BLUR-YOLO, which incorporates techniques to enhance UAV aerial image target detection. The model performs well on blurred images, demonstrating robustness in practical UAV scenarios.

Li et al. [33] presented FM-YOLOv8, a lightweight model tailored for gesture recognition. By optimizing YOLOv8's architecture, the model achieves high accuracy while maintaining computational efficiency. This work underscores the versatility of YOLO models across various applications.

Hanif et al. [34] extended lightweight detection frameworks by developing a model specifically for autonomous vehicles in mining environments. The approach focuses on efficient multi-object detection tailored to challenging conditions. Xu and Wu [35] introduced MRFF-YOLO, a multi-receptive fields fusion network designed for remote sensing target detection. The integration of receptive field fusion techniques ensures accurate detection in remote sensing scenarios. Tang et al. [36] proposed YOLO-RSFM, an efficient road small object detection method that incorporates innovative strategies for accurate detection in road scenarios. Similarly, Chen et al. [37] proposed TO-YOLOX, an advanced tiny object detection model optimized for remote sensing imagery.

2.3 Applications in UAV and Remote Sensing

Pioneering work in UAV-based detection was introduced by Zhu et al. [38], who highlighted challenges and solutions in drone object detection through the VisDrone-DET2018 benchmark. This benchmark provided a comprehensive dataset and evaluation metrics,

enabling researchers to develop and compare detection methodologies tailored for UAV scenarios. Their contributions laid the foundation for advancing UAV detection benchmarks and methodologies.

Ali and Anwar [39] proposed a small target detection benchmark for aerial imagery, addressing the unique challenges of aerial detection in real-world scenarios. Shi et al. [40] developed ASG-YOLOv5, an improved YOLOv5 variant tailored for UAV scenarios. This model incorporates spatial gating and attention mechanisms to enhance small object detection in remote sensing tasks. Liu et al. [41] proposed YOLO-Extract, which enhances YOLOv5 for detecting aircraft in remote sensing imagery, achieving significant performance gains. Qiu et al. [42] developed YOLO-GNS, a UAV perspective-based special vehicle detection network. This model highlights the adaptability of YOLO variants to UAV-specific challenges. Additionally, Li et al. [43] introduced RSI-YOLO, which integrates improved YOLO techniques to optimize remote sensing object detection. Lastly, Sowmya and Radha [44] demonstrated the effectiveness of YOLOv4 for heavy vehicle detection, employing data augmentation and transfer learning to enhance detection capabilities in various settings.

2.4 Limitations of Existing Adverse-Weather Detection Methods

Despite advancements in deraining, defogging, domain adaptation, and attention-augmented YOLO variants, existing research shows four core limitations:

1. Lack of unified robustness across multiple weather types. Most models specialize in a single condition (rain, haze, snow, or low-light) and fail when distortions overlap.
2. Inconsistent backbone and attention integration. Generative VLM datasets and DETR-based transformers improve context understanding but do not provide lightweight fusion suitable for real-time or embedded deployment.
3. Loss of feature stability under visibility collapse. YOLO variants enhance local texture extraction but struggle to retain global structure when haze blur or snow occlusion suppress edge and contour information.
4. High computational overhead. Transformer and DETR-based detectors often exceed 70–150 GFLOPs, preventing real-time usage in UAV and autonomous driving systems with strict latency constraints.

These issues are consistent with robustness challenges observed in recent dual-attention and layer-frozen network designs, where performance degrades whenever local texture cues and global semantic context are not jointly preserved.

Table 1 consolidates prior approaches, including generative vision-language models, YOLO-based weather adaptation, lightweight mobile variants, and DETR-based global reasoning. The comparison confirms that no existing method simultaneously provides: (i) weather-aware global structure recovery, (ii) cross-scale feature stability, and (iii) real-time computation below 15 GFLOPs.

This identifies the architectural motivation for YOLOv8s-WAMNet, which integrates EHViT for structure restoration, CDMSA for scale interaction stability, and CMAF for context-guided feature fusion.

3 Proposed Methodology

This section introduces the proposed YOLOv8s-WAMNet architecture, which enhances vehicle detection robustness under adverse weather conditions while maintaining real-time efficiency. The framework extends the YOLOv8s baseline through targeted architectural refinements in the backbone, neck, detection head, and loss function.

YOLOv8s-WAMNet employs an Efficient Hybrid Vision Transformer backbone with an integrated Cross-Dimensional Multi-Scale Attention mechanism to improve weather-robust feature extraction by jointly modeling local and global context. A Contextual Multi-Attention Fusion module is introduced in the neck to adaptively aggregate multi-scale features and emphasize object-relevant regions. The detection head incorporates a Multi-Head Dynamic Attention strategy to enhance scale-aware localization and decoding accuracy. In addition, a hybrid SIoU-MPDIoU loss is adopted to improve bounding-box regression under complex visual distortions.

The overall architecture and component interactions are illustrated in Fig. 4.

3.1 Efficient Hybrid Vision Transformer (EHViT)

Adverse weather conditions such as fog, rain, snow, and low illumination cause non-uniform visibility degradation, contrast attenuation, and structural ambiguity, which significantly degrade the feature extraction capability of conventional CNN and transformer backbones.

To address these challenges, we propose the Efficient Hybrid Vision Transformer (EHViT) as the backbone of YOLOv8s-WAMNet, as illustrated in Fig. 1. EHViT is designed to jointly preserve local structural details, capture global contextual dependencies, and maintain computational efficiency under adverse weather scenarios.

Fig. 1 depicts the internal structure of a single EHViT block. In the overall backbone, this block is instantiated repeatedly at four hierarchical stages corresponding to strides $\{4, 8, 16, 32\}$, following the pyramidal design of YOLOv8s. Each instantiation operates independently on the feature map produced by the preceding backbone layer and preserves its spatial resolution and channel dimensionality.

At each stage, EHViT processes the input feature map through two parallel branches. The local branch employs a GhostConv stem followed by stacked GhostBottleneck blocks to extract fine-grained structural information such as edges, contours, and small object details while minimizing parameter redundancy. This branch enhances robustness against blur, rain streaks, and low-contrast regions commonly observed in adverse weather.

In parallel, the global branch converts the same feature map into overlapping patch tokens and applies a lightweight transformer encoder to model long-range spatial dependencies and global contextual relationships. Overlapping patch embedding [45] is employed to preserve spatial continuity and fine-grained details that are often lost in non-overlapping tokenization schemes. This property is particularly important for detecting small and partially occluded vehicles, as formulated in Eq. (1).

For an input feature map $I \in \mathbb{R}^{H \times W \times C}$, the overlapping patch embedding is defined as

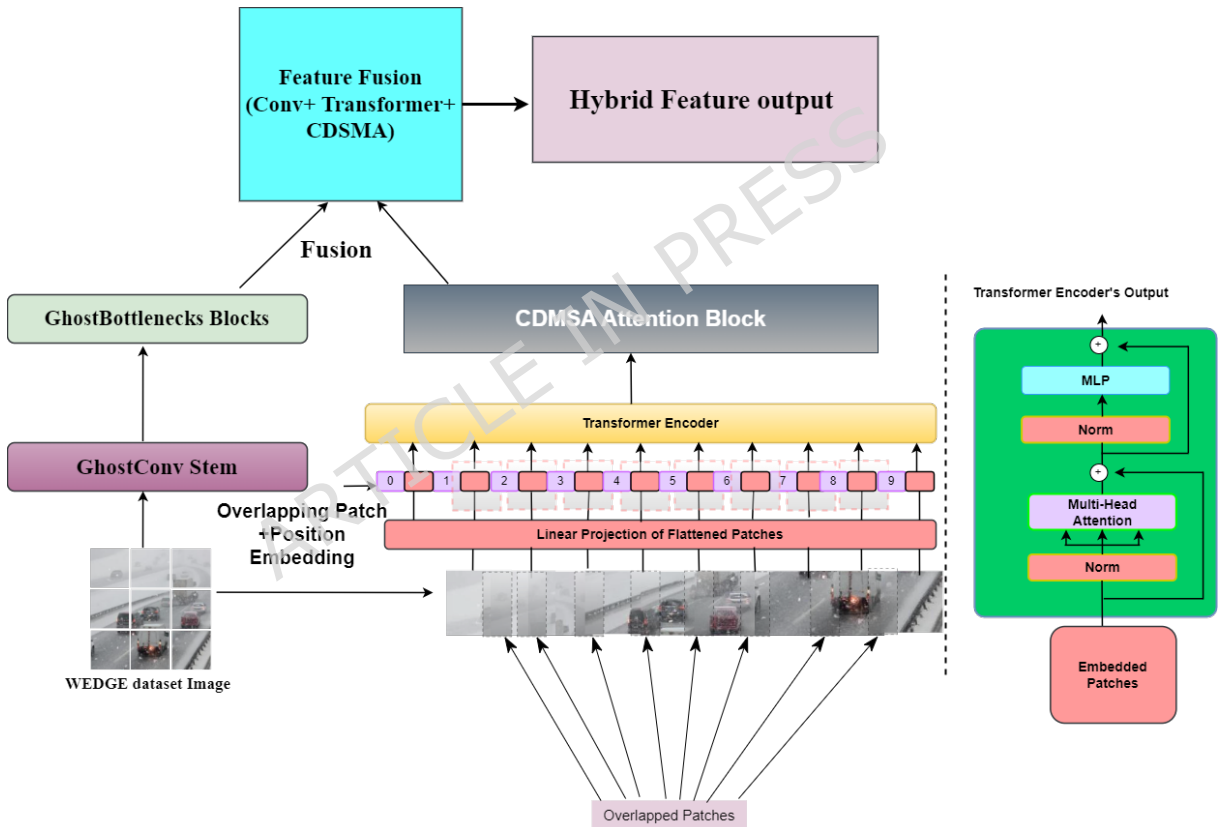
$$\begin{aligned} P_{i,j} &= \phi(I_{i:i+p, j:j+p}), \\ i &= 1, s, \dots, H - p + 1, \\ j &= 1, s, \dots, W - p + 1, \end{aligned} \quad (1)$$

where ϕ is a convolutional embedding, p is the patch size, and $s < p$ is the stride. The overlap $(p - s)$ enhances robustness under adverse weather.

The transformer output is subsequently refined using the Cross-Dimensional Multi-Scale Attention (CDMSA) module, which jointly reweights channel-wise semantics and spatial responses. CDMSA emphasizes informative channels and regions while suppressing weather-induced noise and unreliable activations,

Table 1: Verified quantitative comparison of adverse-weather, lightweight, and transformer-based detection methods.

Category	Model (Ref.)	Params / GFLOPs	mAP@50 (%)	Dataset	Key Limitation
Restoration + Detection	Joint Restoration Pipelines [27, 28]	Not reported	50.69	Snowy datasets	Two-stage pipeline; not real-time; lacks a unified mAP@50 protocol.
Weather-Aware YOLO	ADWNet [29]	13.19M / 13.48	52.30	WEDGE	Vision-only design; limited robustness under haze and low illumination.
Lightweight YOLO	TRC-YOLO [30]	17.8M / 2.98	66.4	PASCAL VOC	Speed-optimized; not evaluated under adverse weather conditions.
Lightweight YOLO	TSD-YOLO [31]	8.8M / 65.6	86.5	CCTSDb	No fog, rain, or snow modeling; lacks cross-weather validation.
Lightweight YOLO	FM-YOLOv8 [33]	1.76M / 5.1	96.6	HaGRID	Gesture-specific task; not designed for general object detection.
DETR-Based	DETR [48]	41M / 86	50.10	COCO	High computational cost; no adverse-weather evaluation reported.
DETR-Based	Deformable DETR [53]	40M / 173	51.30	COCO	High accuracy achieved at the cost of very high GFLOPs.
DETR-Based	DN-DETR [55]	44M / 94	51.91	COCO	Improved convergence; limited robustness under visibility degradation.
DETR-Based	RT-DETR [56]	42M / 136	49.80	COCO	Real-time oriented but computationally heavy for edge deployment.

**Fig. 1:** Overview of the Efficient Hybrid Vision Transformer (EHViT) backbone. The figure illustrates a single EHViT block that is reused at multiple backbone stages with different spatial resolutions and channel dimensions.

thereby strengthening global–local feature interaction without introducing dense attention matrices.

Finally, the refined transformer features are fused with the convolutional features using residual hybrid

fusion, producing a single hybrid feature map per EHViT instantiation. Importantly, EHViT does not change the spatial resolution or channel width of its input; instead, it enhances feature quality at the same scale.

Specifically, at stride 4 (P2) in Table 2, EHViT processes a 160×160 feature map with 128 channels using a single transformer layer with four attention heads, yielding hybrid feature F_2 . At stride 8 (P3), it operates on an $80 \times 80 \times 256$ feature map with one transformer layer and eight attention heads to produce F_3 . At stride 16 (P4), EHViT processes a $40 \times 40 \times 512$ feature map with eight attention heads, generating F_4 . At the deepest stage (stride 32, P5), EHViT processes a $20 \times 20 \times 1024$ feature map using sixteen attention heads, producing F_5 .

As a result, four hybrid feature maps $\{F_2, F_3, F_4, F_5\}$ are generated across the backbone, each corresponding to a different spatial resolution and semantic level. These hybrid features integrate Ghost-based convolutional locality with transformer-based global contextual awareness. At the deepest stage, the F_5 feature is further refined by the SPPF module (Idx 9) to enlarge the receptive field without altering spatial resolution, and the resulting multi-scale features are subsequently forwarded to the neck for feature aggregation and detection.

Compared with pure CNN backbones, EHViT compensates for weather-degraded local textures through global contextual reasoning. Compared with pure transformer backbones, it retains fine-grained spatial detail and efficiency by restricting transformer depth and selectively applying attention at key stages. Overall, EHViT provides a balanced, efficient, and weather-adaptive backbone that significantly improves small-object detection, occlusion handling, and feature stability under adverse weather conditions while maintaining real-time performance.

3.1.1 Cross-Dimensional Multi-Scale Attention (CDMSA)

The Cross-Dimensional Multi-Scale Attention (CDMSA) module (Fig. 2) constitutes a central component of YOLOv8s-WAMNet, aiming to strengthen feature discrimination under adverse weather conditions. Building upon the Efficient Multi-Scale Attention (EMA) mechanism [46], CDMSA integrates lightweight GhostBottleneck-based multi-scale feature encoding with cross-dimensional attention aggregation to improve robustness against visibility degradation.

Let the input feature map be denoted as $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$. The channel dimension is first divided into G groups, where each group $\mathbf{X}_g \in \mathbb{R}^{C_g \times H \times W}$ and $C_g = C/G$. Each group is processed independently using GhostBottleneck blocks with different receptive fields (1×1 , 3×3 , and 5×5) to capture complementary spatial information. The resulting multi-scale features are concatenated along the channel dimension, yielding the aggregated representation $\mathbf{X}_{\text{concat}}$, as defined in Eq. (2).

$$\mathbf{X}_{\text{concat}} = \text{Concat}(\mathbf{X}_{gb1}, \mathbf{X}_{gb2}, \mathbf{X}_{gb3}) \in \mathbb{R}^{3C_g \times H \times W}, \quad (2)$$

Cross-Dimensional Attention Learning.

Based on the aggregated multi-scale feature map $\mathbf{X}_{\text{concat}}$, global channel-wise and spatial-wise descriptors are first extracted to capture complementary cross-dimensional cues. Channel attention is obtained via spatial global average pooling, as defined in Eq. (3), whereas spatial attention is computed through channel global average pooling, as expressed in Eq. (4).

$$\mathbf{A}_{ch}^{raw}(c) = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W \mathbf{X}_{\text{concat}}(c, h, w), \quad (3)$$

$$\mathbf{A}_{sp}^{raw}(h, w) = \frac{1}{C} \sum_{c=1}^C \mathbf{X}_{\text{concat}}(c, h, w). \quad (4)$$

Following the architecture illustrated in Fig. 2, cross-dimensional interaction is performed asymmetrically. In the first branch, the raw channel attention $\mathbf{A}_{ch}^{raw}$ is modulated by the normalized and Softmax-activated spatial attention. Specifically, spatial attention is first normalized and passed through a Softmax function, and the resulting map is then applied to the raw channel attention via element-wise multiplication:

$$\mathbf{A}_{ch} = \mathbf{A}_{ch}^{raw} \odot \text{Softmax}(\text{Norm}(\mathbf{A}_{sp}^{raw})). \quad (5)$$

Conversely, in the second branch, the raw spatial attention $\mathbf{A}_{sp}^{raw}$ is modulated using the normalized channel attention. The channel attention is first normalized, then multiplied element-wise with the raw spatial attention, followed by Softmax activation to obtain the refined spatial attention map:

$$\mathbf{A}_{sp} = \text{Softmax}(\text{Norm}(\mathbf{A}_{ch}^{raw}) \odot \mathbf{A}_{sp}^{raw}). \quad (6)$$

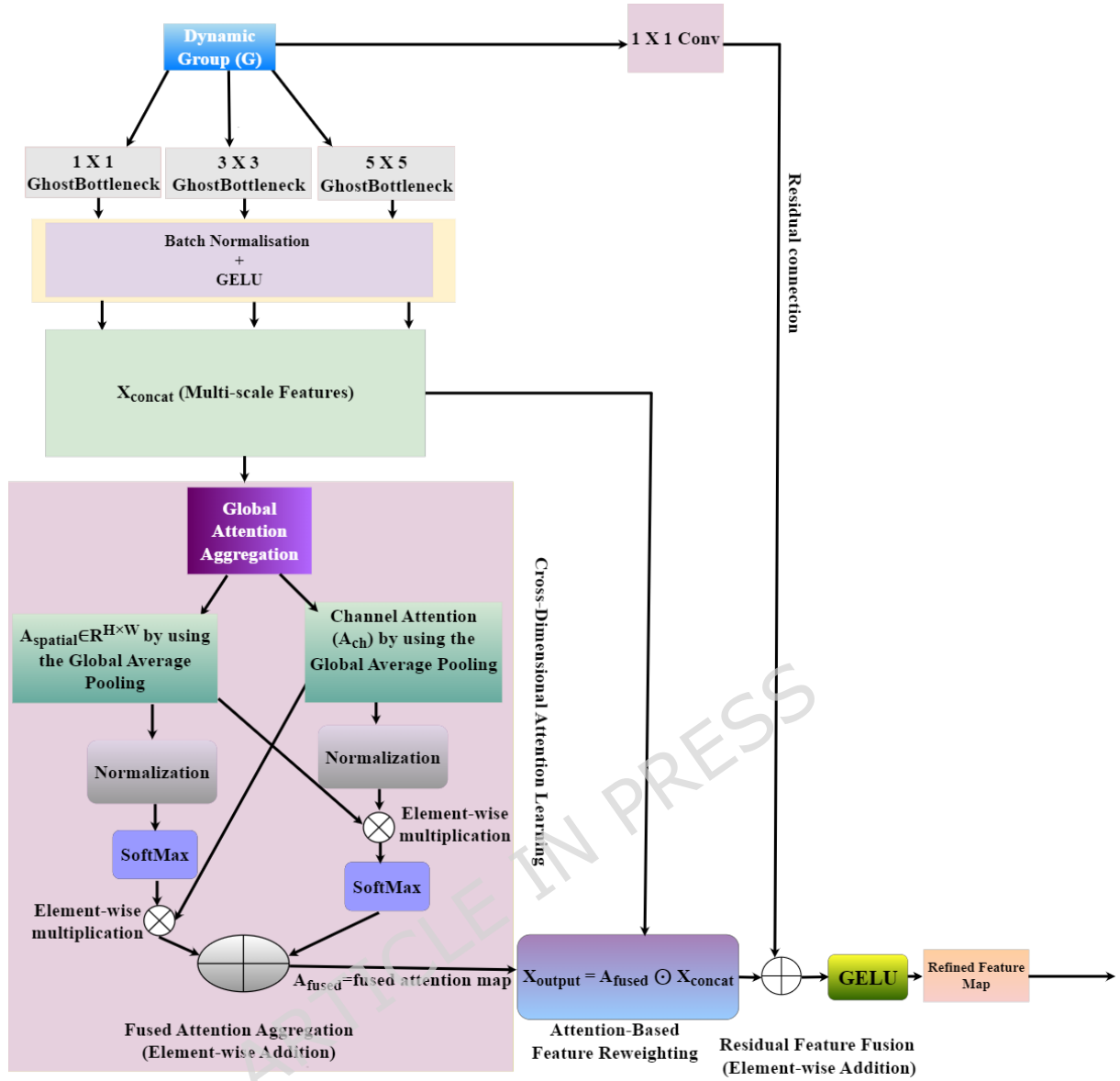


Fig. 2: Structure of the Cross-Dimensional Multi-Scale Attention (CDMSA) Module

This asymmetric cross-dimensional learning strategy enables mutual reinforcement between channel importance and spatial saliency while preserving the directional information flow depicted in the CDMSA architecture.

Fused Attention Aggregation.

The refined channel and spatial attention maps are subsequently aggregated to form a unified attention representation. As defined in Eq. (7), element-wise addition is employed to combine A_{ch} and A_{sp} into the

fused attention map A_{fused} . Element-wise addition is used for attention fusion, whereas element-wise multiplication is reserved exclusively for cross-dimensional interaction and feature reweighting, consistent with the architectural design illustrated in Fig. 2.

$$A_{fused} = A_{ch} + A_{sp}, \quad (7)$$

Feature Reweighting and Residual Learning.

The fused attention map A_{fused} is applied to the multi-scale features through element-wise multiplication to

obtain the attention-weighted feature map $\mathbf{X}_{att}$, as given in Eq. (8). Finally, a residual connection is introduced by adding a 1×1 pointwise projection of the original input feature $\mathbf{X}$, followed by GELU activation, as expressed in Eq. (9). This residual formulation preserves the original feature information while facilitating stable gradient propagation.

$$\mathbf{X}_{att} = \mathbf{A}_{fused} \odot \mathbf{X}_{concat}, \quad (8)$$

$$\mathbf{X}_{out} = \text{GELU}(\mathbf{X}_{att} + \text{Conv}_{1 \times 1}(\mathbf{X})), \quad (9)$$

By jointly leveraging multi-scale GhostBottle-neck encoding, cross-dimensional attention interaction, and residual learning, the CDMSA module enhances global-local contextual representation. This design improves robustness to occlusion, scale variation, and weather-induced feature degradation while maintaining computational efficiency.

3.2 Contextual Multi-Attention Fusion (CMAF) Module

We propose the Contextual Multi-Attention Fusion (CMAF) module (Fig. 3) as a dedicated neck component in YOLOv8s-WAMNet to enhance multi-scale feature consistency and robustness under adverse weather conditions.

CMAF aggregates the EHViT-enhanced backbone features $\{F_2, F_3, F_4\}$ corresponding to strides $\{4, 8, 16\}$, which preserve high-resolution structural cues critical for detecting small- and medium-scale vehicles. The deepest backbone feature F_5 (stride 32) is intentionally excluded from CMAF to avoid semantic dominance and is preserved for subsequent pyramid fusion and detection.

Since F_2 , F_3 , and F_4 have heterogeneous spatial resolutions and channel dimensions, namely $F_2 \in \mathbb{R}^{160 \times 160 \times 128}$, $F_3 \in \mathbb{R}^{80 \times 80 \times 256}$, and $F_4 \in \mathbb{R}^{40 \times 40 \times 512}$, they are first projected into a unified tensor space prior to fusion. Specifically, F_2 is downsampled by a factor of four, F_3 by a factor of two, and F_4 is kept unchanged, followed by 1×1 convolutions for channel alignment, as defined in Eqs. (10)–(12).

$$\hat{F}_2 = \text{Conv}_{1 \times 1}(\text{Down}_4(F_2)), \quad (10)$$

$$\hat{F}_3 = \text{Conv}_{1 \times 1}(\text{Down}_2(F_3)), \quad (11)$$

$$\hat{F}_4 = \text{Conv}_{1 \times 1}(F_4). \quad (12)$$

All aligned features satisfy Eq. (13), enabling direct multi-scale interaction and weighted fusion.

$$\hat{F}_i \in \mathbb{R}^{40 \times 40 \times 512}. \quad (13)$$

Adaptive Dense Contextual Fusion

The aligned features are merged through adaptive weighted summation to form a single fused representation, as expressed in Eq. (14).

$$F_{\text{fusion}} = \sum_{i=2}^4 w_i \hat{F}_i, \quad w_i = \frac{\exp(\alpha_i)}{\sum_{j=2}^4 \exp(\alpha_j)}. \quad (14)$$

Here, α_i are learnable scalars satisfying $\sum_i w_i = 1$, producing a unified tensor $F_{\text{fusion}} \in \mathbb{R}^{40 \times 40 \times 512}$.

Computational complexity. The fusion operation exhibits linear complexity, as given in Eq. (15), since the number of fused scales is fixed ($n = 3$).

$$\mathcal{O}_{\text{fusion}} = \mathcal{O}(C \cdot H \cdot W). \quad (15)$$

Contextual Cross-Attention (CCA)

Inter-channel contextual relationships are modeled on the fused tensor using lightweight attention, as defined in Eq. (16).

$$F_{\text{CCA}} = \text{Attention}(F_{\text{fusion}}). \quad (16)$$

Computational complexity. The CCA operation also maintains linear complexity, as shown in Eq. (17).

$$\mathcal{O}_{\text{CCA}} = \mathcal{O}(C \cdot H \cdot W). \quad (17)$$

Dynamic Efficient Channel Attention (DECA)

Channel saliency is further refined using global channel descriptors and dynamic convolution, as formulated in Eq. (18).

$$F_{\text{avg}} = \frac{1}{HW} \sum_{h,w} F_{\text{CCA}}(h, w), \quad (18)$$

$$F_{\text{DECA}} = F_{\text{CCA}} \odot \sigma(\text{Conv1D}(F_{\text{avg}})).$$

Computational complexity. The DECA module operates with linear complexity, as indicated in Eq. (19).

$$\mathcal{O}_{\text{DECA}} = \mathcal{O}(C \cdot H \cdot W). \quad (19)$$

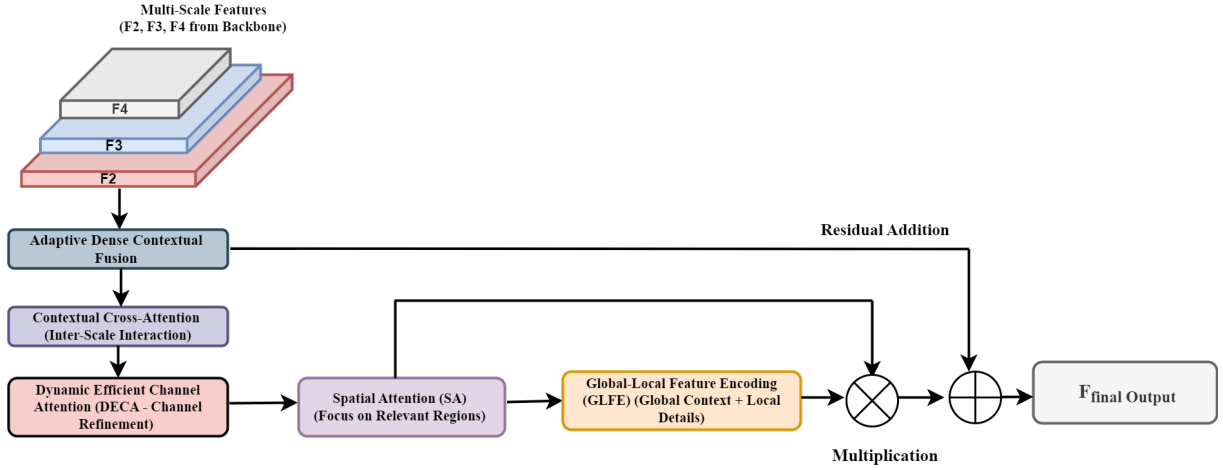


Fig. 3: Block diagram of the proposed Contextual Multi-Attention Fusion (CMAF) module.

Spatial Attention (SA)

Important spatial regions are emphasized using spatial attention, as expressed in Eq. (20).

$$F_{SA} = F_{DECA} \odot \sigma \left(\text{Conv2D} \left(\left[\text{MaxPool}(F_{DECA}), \text{AvgPool}(F_{DECA}) \right] \right) \right). \quad (20)$$

Computational complexity. The spatial attention operation preserves linear complexity, as shown in Eq. (21).

$$\mathcal{O}_{SA} = \mathcal{O}(C \cdot H \cdot W). \quad (21)$$

Global-Local Feature Encoding (GLFE)

Global contextual information is injected through combined pooling operations, as defined in Eq. (22).

$$F_{GLFE} = F_{SA} + \text{Transform}(\text{GAP}(F_{SA}) \oplus \text{GMP}(F_{SA})). \quad (22)$$

Computational complexity. The GLFE module maintains linear complexity, as given in Eq. (23).

$$\mathcal{O}_{GLFE} = \mathcal{O}(C \cdot H \cdot W). \quad (23)$$

Final Output

The final CMAF output is obtained through residual attention fusion, as expressed in Eq. (24).

$$F_{CMAF} = (F_{GLFE} \odot F_{SA}) + \gamma F_{fusion}, \quad (24)$$

where γ is a learnable scalar. The resulting tensor $F_{CMAF} \in \mathbb{R}^{40 \times 40 \times 512}$ is passed to the YOLOv8s feature pyramid.

Overall complexity. By construction, all CMAF components operate with linear complexity, as summarized in Eq. (25).

$$\mathcal{O}_{CMAF} = \mathcal{O}(C \cdot H \cdot W). \quad (25)$$

3.3 Multi-Head Dynamic Attention (MHDA) Head

We propose the Multi-Head Dynamic Attention (MHDA) head as a novel detection head for YOLOv8s-WAMNet, designed to jointly enhance local adaptability, global relational reasoning, and fine-grained detection under adverse weather. Unlike prior works that adopt dynamic convolution [47], transformer decoding [48], or auxiliary detection heads in isolation, MHDA integrates all three mechanisms into a unified framework, enabling robustness to small objects, occlusion, and weather-induced degradation.

3.3.1 Dynamic Convolution

Dynamic convolution adaptively generates input-dependent kernels for precise feature extraction. The kernel weights are computed as shown in Eq. (26):

$$\mathbf{W}_{dyn} = \sigma(\mathbf{F}_{in} \cdot \mathbf{W}_{gen}), \quad (26)$$

where $\mathbf{F}_{in}$ denotes the input feature, $\mathbf{W}_{gen}$ is the weight generation matrix, and $\sigma(\cdot)$ represents the sigmoid activation function. The convolution operation is then performed as expressed in Eq. (27):

$$\mathbf{F}_{out} = \mathbf{W}_{dyn} * \mathbf{F}_{in} + \mathbf{b}, \quad (27)$$

where $*$ indicates convolution and $\mathbf{b}$ is the bias term. This mechanism adaptively adjusts the filtering strength according to object scale and structural variations.

3.3.2 Transformer Decoder Integration

A transformer decoder, inspired by DETR [48], is incorporated to capture long-range dependencies and semantic relations among detected objects. The query update at time step t is defined in Eq. (28):

$$\mathbf{Q}_{t+1} = \text{Softmax}\left(\frac{\mathbf{Q}_t \mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V}, \quad (28)$$

where $\mathbf{Q}_t$ represents the query, and $\mathbf{K}$ and $\mathbf{V}$ denote the key and value matrices, respectively. This structure enhances contextual reasoning in dense or cluttered traffic scenarios by modeling global relationships.

3.3.3 Auxiliary Fine-Grained Detection (AFGD) Heads

To strengthen detection performance for small and weather-affected vehicles, auxiliary fine-grained detection heads are employed to process multi-scale feature maps, as shown in Eq. (29):

$$\mathbf{D}_i = \text{Conv}(\mathbf{F}_i), \quad i = 1, 2, \dots, N, \quad (29)$$

and the aggregated output is obtained using Eq. (30):

$$\mathbf{D}_{aux} = \sum_{i=1}^N \alpha_i \mathbf{D}_i, \quad (30)$$

where α_i are learnable weights. These auxiliary heads adaptively emphasize fine-scale and weather-specific cues, ensuring robustness under challenging conditions such as fog, rain, and low illumination.

3.4 Hybrid Shape-aware Intersection over Union–Minimum Point Distance Intersection over Union(SIoU–MPDIoU) Loss

IoU-based regression losses, such as DIoU and CIoU [49], have been widely adopted to improve bounding-box localization accuracy by incorporating geometric constraints beyond pure overlap. Building on this line of research, we propose a hybrid loss formulation that integrates enhanced Shape-aware Intersection over Union (SIoU) and enhanced Minimum Point Distance Intersection over Union (MPDIoU) to jointly optimize center distance, shape consistency, angular alignment, and aspect-ratio stability.

Enhanced SIoU

The proposed loss extends SIoU [14] by introducing adaptive shape and angular penalties. The enhanced SIoU formulation is defined in Eq. (31),

$$L_{\text{SIoU}}^E = 1 - \text{IoU} + \frac{\Delta + \Omega}{2} + \alpha_{\text{ang}} \Lambda, \quad (31)$$

where Δ denotes the center-distance term and Ω represents the shape-consistency term. For numerical robustness, an arctangent-based angular penalty is employed, as expressed in Eq. (32),

$$\Lambda = \left(\frac{\Delta\theta}{\pi/2}\right)^2, \quad \Delta\theta = \left|\arctan \frac{h_p}{w_p} - \arctan \frac{h_g}{w_g}\right|. \quad (32)$$

The angular weighting factor is adaptively modulated according to shape consistency, as defined in Eq. (33),

$$\alpha_{\text{ang}} = \frac{\text{IoU}}{\text{IoU} + \exp(-\delta \Omega)}, \quad (33)$$

where $\delta > 0$ is a tuning parameter.

Enhanced MPDIoU

The MPDIoU loss [49, 50] is further enhanced by incorporating normalized corner distance, aspect-ratio consistency, and IoU-adaptive angular scaling. The enhanced MPDIoU formulation is given in Eq. (34),

$$L_{\text{MPDIoU}}^E = 1 - \text{IoU} + \lambda_1 \tilde{d}_{\min} + \lambda_2 v + \beta d_{\text{angle}}, \quad (34)$$

where the individual terms are defined in Eqs. (35)–(37),

$$\tilde{d}_{\min} = \frac{\min_i \sqrt{(x_{i,p} - x_{i,g})^2 + (y_{i,p} - y_{i,g})^2}}{\max(w_p, h_p)}, \quad (35)$$

$$v = \left(\frac{w_p}{h_p} - \frac{w_g}{h_g} \right)^2, \quad (36)$$

$$d_{\text{angle}} = \left(\arctan \frac{h_p}{w_p} - \arctan \frac{h_g}{w_g} \right)^2. \quad (37)$$

The angular scaling factor β is adaptively determined by the IoU value, as defined in Eq. (38),

$$\beta = \frac{1}{1 + \exp(-\kappa \text{IoU})}. \quad (38)$$

Hybrid Loss

The final regression objective combines the enhanced SIoU and MPDIoU losses using a dynamic hybrid weighting strategy. The hybrid loss is defined in Eq. (39),

$$L_{\text{Hybrid}} = \alpha_{\text{hyb}} L_{\text{SIoU}}^E + (1 - \alpha_{\text{hyb}}) L_{\text{MPD}}^E, \quad (39)$$

where the adaptive weighting factor α_{hyb} is determined by Eq. (40),

$$\alpha_{\text{hyb}} = \frac{\text{IoU}}{\text{IoU} + \exp(-\eta v)}, \quad (40)$$

and $\eta > 0$ controls the sensitivity to aspect-ratio inconsistency v .

3.5 YOLOv8s-WAMNet YAML Architecture

The complete architectural configuration of the proposed YOLOv8s-WAMNet is summarized in Table 2 and illustrated in Fig. 4. The network follows the standard YOLOv8s backbone–neck–head paradigm to preserve tensor compatibility, stable gradient propagation, and training reliability, while selectively integrating attention-enhanced modules to improve robustness under adverse weather conditions. In accordance with YOLOv8 scaling rules, a width multiplier w is used throughout the architecture to uniformly scale channel dimensions across all layers.

In the backbone, GhostConv layers (Idx 0, Idx 1, Idx 3, Idx 5, and Idx 7) perform progressive downsampling using 3×3 kernels with stride 2, producing feature maps at spatial resolutions 320×320 , $160 \times$

160 , 80×80 , 40×40 , and 20×20 , respectively. After width scaling, these stages produce channel dimensions $\{64w, 128w, 256w, 512w, 1024w\}$. EHViT blocks with integrated CDMSA (Idx 2, Idx 4, Idx 6, and Idx 8) are inserted after each major downsampling stage and refine features while preserving spatial resolution and channel dimensionality. These stages generate four hybrid backbone feature maps $\{F_2, F_3, F_4, F_5\}$ corresponding to strides $\{4, 8, 16, 32\}$ with tensor dimensions $F_2 \in \mathbb{R}^{160 \times 160 \times 128w}$, $F_3 \in \mathbb{R}^{80 \times 80 \times 256w}$, $F_4 \in \mathbb{R}^{40 \times 40 \times 512w}$, and $F_5 \in \mathbb{R}^{20 \times 20 \times 1024w}$. Each F_i integrates Ghost-based convolutional local representations with transformer-based global contextual modeling through CDMSA. The SPPF module (Idx 9) is subsequently applied to F_5 to enlarge the receptive field without altering spatial resolution or channel dimensionality.

In the neck, the proposed CMAF module (Idx 10) aggregates the multi-scale backbone features $\{F_2, F_3, F_4\}$ at spatial resolutions 160×160 , 80×80 , and 40×40 , respectively. These features are spatially aligned to the 40×40 scale and fused via adaptive contextual attention to produce a single enhanced feature map $F_{\text{CMAF}} \in \mathbb{R}^{40 \times 40 \times 512w}$, which is denoted as C_4 in the feature pyramid. This tensor replaces the original backbone feature F_4 and serves as the sole CMAF output for subsequent neck operations.

Following the standard YOLOv8s top–down FPN pathway, the deepest backbone feature F_5 is upsampled (Idx 11) and concatenated with F_{CMAF} at Idx 12. The resulting tensor is refined by a C2f block (Idx 13) to produce the intermediate feature $T_4 \in \mathbb{R}^{40 \times 40 \times 512w}$. This feature is then upsampled (Idx 14) and concatenated with F_3 (Idx 15), after which another C2f block (Idx 16) generates the high-resolution feature $T_3 \in \mathbb{R}^{80 \times 80 \times 256w}$ for small-object detection.

A bottom–up PAN pathway is subsequently constructed. Feature T_3 is downsampled (Idx 17) and concatenated with T_4 (Idx 18), followed by a C2f block (Idx 19) to produce the medium-scale feature $P_4 \in \mathbb{R}^{40 \times 40 \times 512w}$. This feature is further downsampled (Idx 20) and concatenated with F_5 (Idx 21), and a final C2f block (Idx 22) produces the large-scale feature $P_5 \in \mathbb{R}^{20 \times 20 \times 1024w}$.

Finally, the MHDA detection head (Idx 23) operates on $\{T_3, P_4, P_5\}$, corresponding to spatial resolutions 80×80 , 40×40 , and 20×20 , respectively. The argument $[16, 19, 22, nc]$ specifies the indices of these feature maps and the number of target classes nc , enabling multi-scale, attention-enhanced object detection across small, medium, and large objects.

Table 2: Architecture configuration of YOLOv8s-WAMNet with explicit tensor resolution and channel dimensionality. F_i denotes backbone feature maps. CMAF aggregates contextual information from $\{F_2, F_3, F_4\}$ and produces F_{CMAF} (denoted as C_4), which replaces F_4 in the FPN stage.

Part	Idx	Module	Arguments
Backbone	0	GhostConv	$[640 \times 640 \times 3 \rightarrow 320 \times 320 \times 64, k=3, s=2]$
	1	GhostConv	$[320 \times 320 \times 64 \rightarrow 160 \times 160 \times 128, k=3, s=2]$
	2	EHViT+CDMSA	$[F_2 : 160 \times 160 \times 128, 1\text{-layer}, 4\text{-heads}]$
	3	GhostConv	$[160 \times 160 \times 128 \rightarrow 80 \times 80 \times 256, k=3, s=2]$
	4	EHViT+CDMSA	$[F_3 : 80 \times 80 \times 256, 1\text{-layer}, 8\text{-heads}]$
	5	GhostConv	$[80 \times 80 \times 256 \rightarrow 40 \times 40 \times 512, k=3, s=2]$
	6	EHViT+CDMSA	$[F_4 : 40 \times 40 \times 512, 1\text{-layer}, 8\text{-heads}]$
	7	GhostConv	$[40 \times 40 \times 512 \rightarrow 20 \times 20 \times 1024, k=3, s=2]$
	8	EHViT+CDMSA	$[F_5 : 20 \times 20 \times 1024, 1\text{-layer}, 16\text{-heads}]$
Neck + Head	9	SPPF	$[20 \times 20 \times 1024 \rightarrow 20 \times 20 \times 1024]$
	10	CMAF	$[\{F_2, F_3, F_4\} \rightarrow F_{\text{CMAF}} (= C_4) : 40 \times 40 \times 512]$
	11	Upsample	$[F_5 : 20 \times 20 \times 1024 \rightarrow 40 \times 40 \times 1024]$
	12	Concat	$[[11, F_{\text{CMAF}}] \rightarrow 40 \times 40 \times (1024 + 512)]$
	13	C2f	$[40 \times 40 \times 1536 \rightarrow T_4 : 40 \times 40 \times 512]$
	14	Upsample	$[T_4 : 40 \times 40 \times 512 \rightarrow 80 \times 80 \times 512]$
	15	Concat	$[[14, F_3] \rightarrow 80 \times 80 \times (512 + 256)]$
	16	C2f	$[80 \times 80 \times 768 \rightarrow T_3 : 80 \times 80 \times 256]$
	17	Conv	$[T_3 : 80 \times 80 \times 256 \rightarrow 40 \times 40 \times 256, k=3, s=2]$
	18	Concat	$[[17, T_4] \rightarrow 40 \times 40 \times (256 + 512)]$
	19	C2f	$[40 \times 40 \times 768 \rightarrow P_4 : 40 \times 40 \times 512]$
	20	Conv	$[P_4 : 40 \times 40 \times 512 \rightarrow 20 \times 20 \times 512, k=3, s=2]$
	21	Concat	$[[20, F_5] \rightarrow 20 \times 20 \times (512 + 1024)]$
	22	C2f	$[20 \times 20 \times 1536 \rightarrow P_5 : 20 \times 20 \times 1024]$
	23	MHDA-Head	$[\{T_3 (80 \times 80 \times 256), P_4 (40 \times 40 \times 512), P_5 (20 \times 20 \times 1024)\}, nc]$
Scale hyper-parameters: depth = 0.33, width = 0.50, max channels = 1024			

4 Experimental Setup and Results

4.1 Dataset

For reliable autonomous driving, object detection models must perform robustly under diverse weather conditions. To train and evaluate the proposed YOLOv8s-WAMNet, both synthetic and real-world adverse-weather datasets were employed. Specifically, the WEDGE (Weather images by DALL-E Generation) dataset [26] and the DAWN (Detection in Adverse Weather Nature) dataset [51] were used to ensure comprehensive evaluation across controlled and naturally occurring weather conditions.

4.1.1 WEDGE Dataset Overview

The WEDGE dataset comprises 3,360 images annotated for vehicle detection across 16 weather types, including snow, rain, fog, dust, hurricane, and night scenes. It contains seven object classes: car, truck, bus, motorcycle, bicycle, person, and van. The category-wise distribution of bounding boxes is illustrated in Fig. 5.

The dataset is divided into training, validation, and testing subsets as summarized in Table 3.

Table 3: Summary of WEDGE Dataset Splits

Split	Images	Boxes	Share (%)
Training	2,351	10,696	70.0
Validation	672	3,408	20.0
Testing	337	1,737	10.0
Total	3,360	16,513	100.0

4.1.2 DAWN Dataset Overview

To assess the real-world generalization ability of the proposed approach, we utilize the DAWN (Detection in Adverse Weather Nature) dataset [51], which comprises traffic scenes captured under naturally occurring adverse weather conditions.

The DAWN dataset contains 1,026 real-world images acquired under four challenging weather scenarios: fog (300 images), snow (204 images), rain (200

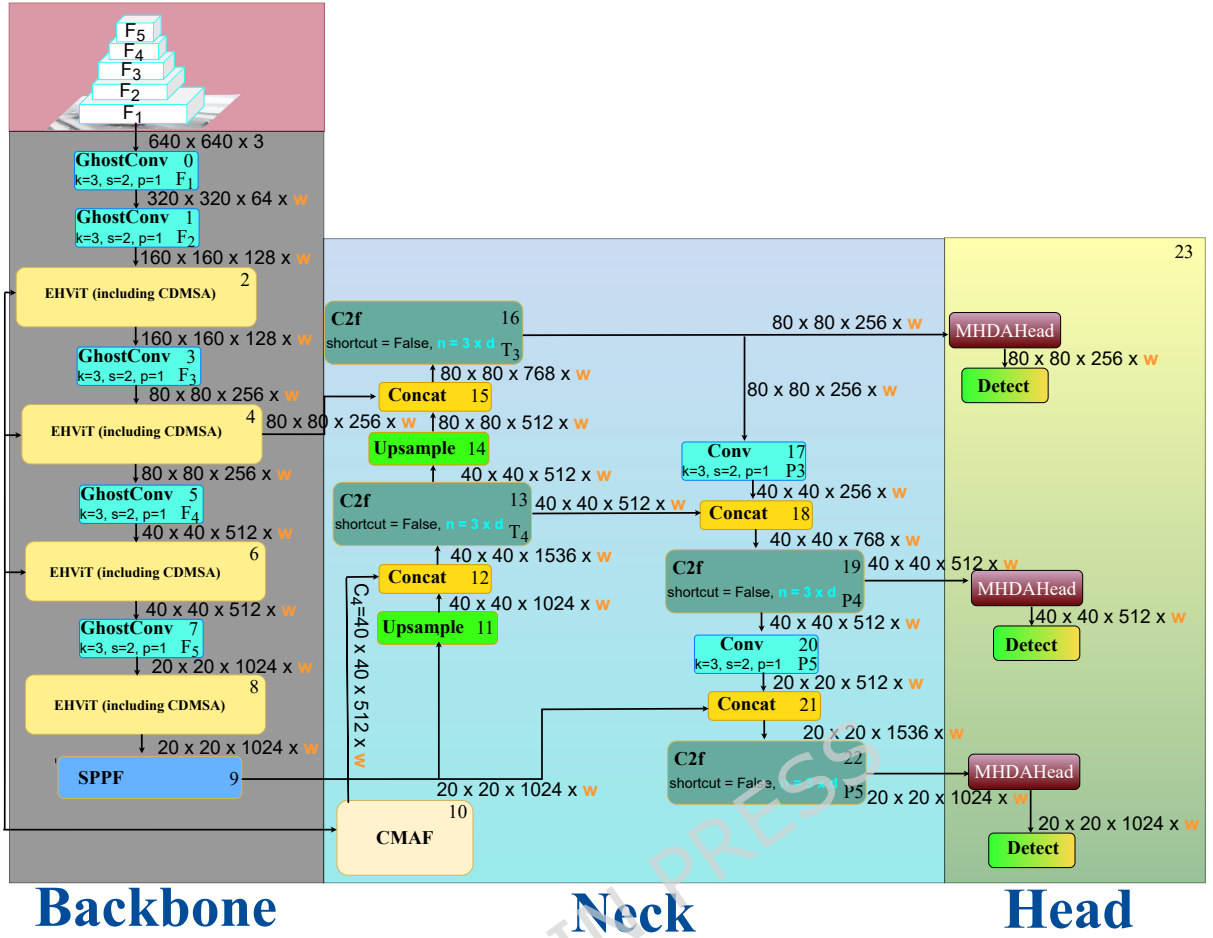


Fig. 4: Architecture of YOLOv8s-WAMNet

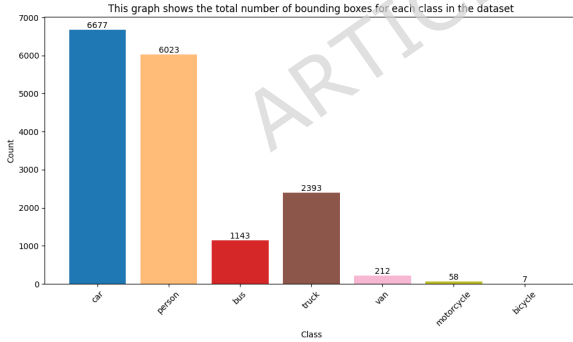


Fig. 5: Distribution of object categories and bounding box counts in the WEDGE dataset.

images), and sandstorms (323 images). These adverse conditions lead to significant visual degradation, including reduced visibility, low contrast, motion blur, and

background clutter, making DAWN a challenging benchmark for object detection.

All images are annotated with bounding boxes for six object categories: *bicycle*, *bus*, *car*, *motorcycle*, *person*, and *truck*, corresponding to class indices 0–5, respectively. Only samples with valid YOLO-format annotations were retained during the data preparation process.

For experimental evaluation, the dataset is divided into 700 training images, 203 validation images, and 100 testing images. Data augmentation is applied exclusively to the training subset to enhance robustness while preserving the intrinsic characteristics of adverse weather conditions.

The DAWN dataset is employed solely to evaluate the robustness and cross-dataset generalization capability of the proposed YOLOv8s-WAMNet model.

4.2 Simulation Details

All experiments were conducted on a high-performance computing setup equipped with an NVIDIA A100 GPU featuring 80 GB of VRAM and 6,912 CUDA cores, supported by 80 GB of system RAM and a 2 TB SSD for fast data access. The system operated on Ubuntu 22.04 with Python 3.10 and PyTorch 2.4.1 configured with CUDA 12.1. Key supporting libraries included NumPy, OpenCV, Albumentations, Matplotlib, Pandas, and TorchVision. For optimized inference, TensorRT and ONNX Runtime frameworks were employed.

All YOLOv-based models were trained and evaluated under the same and equal experimental conditions, including a fixed input image resolution of 640×640 pixels, identical dataset splits, and an identical computational environment on both the WEDGE and DAWN datasets. Under this unified setup, the proposed YOLOv8s-WAMNet and the YOLOv8s baseline were trained using the same hyperparameter configuration—covering batch size, optimizer settings, learning rate scheduling, data augmentation strategy, and number of training epochs—to ensure a fair and controlled comparison.

For transformer-based detectors, including DETR, Deformable DETR, Conditional DETR, and DN-DETR, training was performed under the same environment and input resolution while adopting architecture-specific optimization settings and loss formulations as recommended in their original implementations. This strategy preserves each model's intended optimization behavior and avoids performance degradation caused by inappropriate parameter constraints.

Data augmentation was performed using the Albumentations library, applying random blur, CLAHE, horizontal flipping, grayscale conversion, and color adjustments with hue, saturation, and value parameters set to 0.015, 0.7, and 0.4, respectively. Geometric transformations included rotation within $\pm 10^\circ$, scaling by a factor of 0.5, and horizontal flipping with a probability of 50%.

During inference, the confidence threshold was set to 0.40, with an IoU threshold of 0.50. Post-processing employed Non-Maximum Suppression with an IoU threshold of 0.40. The inference pipeline was further optimized using FP16 precision, TensorRT acceleration, and ONNX Runtime to ensure high-throughput evaluation.

4.3 Simulation Results

4.3.1 Model Complexity and Computational Cost

To achieve optimal performance, extensive hyperparameter tuning was carried out on the WEDGE dataset. The optimal configuration for training the proposed YOLOv8s-WAMNet model is presented in Table 4. The configuration includes an input image size of 640×640 , a batch size of 16, and the AdamW optimizer with a learning rate of 0.001, weight decay of 0.0005, and momentum of 0.937. The model was trained for more than 300 epochs using an IoU threshold of 0.5. Data augmentation was applied using the Albumentations library, incorporating blur, CLAHE, horizontal flipping, and grayscale conversion to improve model robustness under diverse weather and lighting conditions.

To evaluate the computational efficiency and scalability of YOLOv8s-WAMNet, we compared it with multiple state-of-the-art object detection architectures, including YOLOv8s [12], YOLOv9s [13], YOLOv10s [13], YOLOv11s [13], YOLOv12s [52], and several Transformer-based detectors such as DETR [48], Deformable DETR [53], Conditional DETR [54], and DN-DETR [55]. The comparison focuses on the number of parameters and GFLOPs, as these are critical indicators of model complexity and real-time deployment feasibility in autonomous systems.

As shown in Table 5, the baseline YOLOv8s comprises 11.13 M parameters and 28.7 GFLOPs. Subsequent YOLO versions progressively reduce computational cost while maintaining detection accuracy: YOLOv9s (7.29 M, 27.4 GFLOPs), YOLOv10s (8.07 M, 24.8 GFLOPs), YOLOv11s (9.43 M, 21.6 GFLOPs), and YOLOv12s (9.09 M, 19.6 GFLOPs). Transformer-based models, such as DETR and its variants, demonstrate strong accuracy but incur high computational loads, with parameter counts exceeding 40 M and GFLOPs above 70.

In contrast, the proposed YOLOv8s-WAMNet achieves an efficient balance between detection accuracy and computational efficiency, requiring only 9.69 M parameters and 13.24 GFLOPs—the lowest among all compared models. Compared with ADWNet [29], which uses 13.19 M parameters and 13.48 GFLOPs, YOLOv8s-WAMNet attains a 26.5% reduction in parameters and a 1.8% decrease in computational cost while maintaining superior detection robustness under adverse weather.

These results validate that the integration of the Hybrid Attention and Multi-Scale Fusion module significantly enhances computational efficiency and detection accuracy, making YOLOv8s-WAMNet highly suitable for real-time vehicle detection in autonomous driving and embedded edge-computing applications.

Table 4: Best Hyperparameter Configuration for YOLOv8s-WAMNet

Hyperparameter	Optimal Value
Input Image Size	640
Batch Size	16
Optimizer	AdamW
Learning Rate	0.001
Weight Decay	0.0005
Momentum	0.937
IoU Threshold	0.5
Data Augmentation	Albumentations (Blur, CLAHE, Flip, ToGray)
Epochs	more than 300

Table 5: Comparison of model complexity and computational cost

Model	Parameters (M)	GFLOPs
YOLOv8s [12]	11.13	28.7
YOLOv9s [13]	7.29	27.4
YOLOv10s [13]	8.07	24.8
YOLOv11s [13]	9.43	21.6
YOLOv12s [52]	9.09	19.6
DETR [48]	41.3	86.0
Deformable DETR [53]	40.1	72.0
Conditional DETR [54]	44.1	90.0
DN-DETR [55]	44.3	85.3
YOLOv8s-WAMNet (Proposed)	9.69	13.24

4.4 Ablation Study and Component Analysis

This section presents a comprehensive ablation analysis to quantify the individual contributions of the proposed architectural modules and the hybrid loss function in YOLOv8s-WAMNet. All experiments were conducted on the WEDGE dataset under identical training and evaluation settings to ensure fair and reproducible comparison. The ablation study is divided into two parts: architectural component ablation, which analyzes the

impact of each network module, and loss function ablation, which independently validates the effectiveness of the proposed hybrid SIoU-MPDIoU regression objective.

To ensure clear module-level contribution attribution, the ablation strategy isolates individual components while keeping the remaining architecture unchanged. In each variant, a single module is introduced or removed while retaining the original YOLOv8s Backbone, Neck, or Head design for all other components. This controlled evaluation framework enables direct attribution of performance gains to specific modules and mitigates the influence of cumulative architectural changes or overfitting.

4.4.1 Architectural Component Ablation

To assess the contribution of individual architectural components in the proposed YOLOv8s-WAMNet framework, a systematic ablation study was conducted on the WEDGE dataset. The analysis progressively integrates the proposed modules into the baseline architecture, enabling a controlled evaluation of their incremental impact on vehicle detection performance under adverse weather conditions.

All ablation variants were trained and evaluated under identical experimental settings. Performance was measured using mean Average Precision at an IoU threshold of 0.5 (mAP@50) and the averaged metric over IoU thresholds ranging from 0.5 to 0.95 (mAP@50–95), ensuring fair, consistent, and reproducible comparisons across different architectural configurations.

The baseline YOLOv8s-WAMNet achieves an initial performance of 45.6% mAP@50 and 22.6% mAP@50–95. Incorporating the Efficient Hybrid Vision Transformer (EHViT) backbone significantly enhances feature representation capability, leading to improved performance of 49.32% mAP@50 and 23.45% mAP@50–95.

Further integration of the Cross-Dimensional Multi-Scale Attention (CDMSA) module strengthens spatial-channel feature interactions under adverse weather conditions, resulting in a notable improvement to 51.79% mAP@50 and 25.14% mAP@50–95. When the Contextual Multi-Attention Fusion (CMAF) module is introduced to aggregate multi-scale backbone features, the model maintains competitive performance, achieving 51.01% mAP@50 and 24.65% mAP@50–95, indicating improved contextual consistency with moderate regularization effects.

Table 6: Progressive and removal-based architectural ablation study of YOLOv8s-WAMNet on the WEDGE dataset

Model Variant	mAP@50 (%)	mAP@50–95 (%)
Baseline (YOLOv8s-WAMNet w/o modifications)	45.6	22.6
+ EHViT Backbone	49.32	23.45
+ EHViT + CDMSA	51.79	25.14
+ EHViT + CDMSA + CMAF	51.01	24.65
+ EHViT + CDMSA + CMAF + MHDA-Head	50.91	23.49
Final YOLOv8s-WAMNet (Proposed)	51.90	24.96

Subsequent incorporation of the Multi-Head Dynamic Attention (MHDA) detection head further refines feature discrimination and localization, yielding 50.91% mAP@50 and 23.49% mAP@50–95. Although this configuration emphasizes adaptive attention in the detection head, the results indicate that balanced integration of backbone, neck, and head components is essential for optimal performance.

The final YOLOv8s-WAMNet configuration achieves the best overall results, with 51.90% mAP@50 and 24.96% mAP@50–95, outperforming all intermediate variants. As summarized in Table 6, these results validate the complementary contributions of EHViT, CDMSA, CMAF, and MHDA-Head, demonstrating their collective effectiveness in enhancing detection accuracy, localization robustness, and resilience to adverse weather conditions.

4.4.2 Loss Function Ablation Study

To independently validate the effectiveness of the proposed hybrid SIoU–MPDIoU loss, we conducted a dedicated loss-function ablation study on the WEDGE dataset. In this experiment, the full YOLOv8s-WAMNet architecture was fixed, and only the bounding-box regression loss was varied to isolate its impact on detection performance.

Table 7: Loss function ablation study on the WEDGE dataset using the full YOLOv8s-WAMNet architecture

Loss Function	mAP@50 (%)	mAP@50–95 (%)
SIoU only	51.12	24.31
MPDIoU only	51.34	24.48
Hybrid SIoU–MPDIoU (Proposed)	51.90	24.96

The ablation results in Table 7 clearly demonstrate the complementary behavior of the two loss components. When used independently, SIoU improves localization by enforcing shape consistency and angular alignment, resulting in a noticeable gain over standard IoU-based regression. MPDIoU further enhances performance by emphasizing minimum point distance and corner-level geometric constraints, yielding slightly higher accuracy than SIoU alone.

However, both losses capture partially overlapping yet incomplete geometric properties when applied in isolation. SIoU is more effective in regulating orientation and aspect ratio consistency, while MPDIoU excels at minimizing spatial misalignment and boundary deviation. The proposed hybrid SIoU–MPDIoU formulation dynamically balances these complementary constraints through IoU-adaptive weighting, enabling more stable convergence and more precise bounding-box regression.

As a result, the hybrid loss achieves the highest performance, improving mAP@50 by 0.56 and 0.78 percentage points over MPDIoU-only and SIoU-only configurations, respectively, and yielding the best mAP@50–95 score. These findings confirm that the hybrid strategy is not redundant but instead essential for maximizing localization accuracy under adverse weather conditions.

4.5 Performance Comparison on the WEDGE Dataset

Table 8 and Table 9 present a comprehensive comparison of object detection models evaluated on the WEDGE dataset [26]. The unified analysis highlights both overall and class-wise detection performance across single-stage YOLO variants, transformer-based DETR architectures, and the proposed YOLOv8s-WAMNet model.

Classical baselines such as Faster R-CNN and MobileNet variants report only a single-value mean

average precision (mAP), with Faster R-CNN (ResNet-50, fine-tuned on WEDGE) achieving 45.41%, while MobileNet-Large and MobileNet-Large-320 record 43.30% and 34.98%, respectively. These results provide baseline performance under multi-weather driving scenarios.

Among single-stage detectors, YOLOv8s achieves $\text{mAP}@50 = 45.60\%$ and $\text{mAP}@50-95 = 22.60\%$. Subsequent YOLO versions show limited improvement: YOLOv9s (38.30%, 17.70%), YOLOv10s (38.80%, 18.70%), YOLOv11s (45.70%, 23.20%), and YOLOv12s (44.60%, 22.12%) [12, 13, 51]. These models maintain real-time performance but exhibit reduced robustness in low-visibility conditions.

Transformer-based models, including DETR and its enhanced variants [48, 53–56], achieve stronger feature representations at higher computational cost. DETR attains $\text{mAP}@50 = 48.50\%$ and $\text{mAP}@50-95 = 23.10\%$, while Deformable DETR reaches 50.20% and 24.20%. Conditional DETR and DN-DETR report 49.70%, 23.80% and 50.80%, 24.30%, respectively.

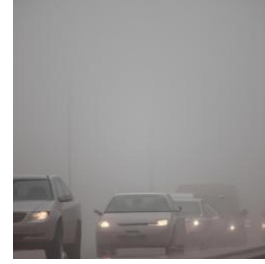
The proposed YOLOv8s-WAMNet achieves the best detection accuracy among all compared models, recording $\text{mAP}@50 = 51.90\%$ and $\text{mAP}@50-95 = 24.96\%$. This represents an absolute improvement of 6.30% in $\text{mAP}@50$ and 2.36% in $\text{mAP}@50-95$ over the YOLOv8s baseline. As shown in Table 9, YOLOv8s-WAMNet also provides superior per-class accuracy, particularly for cars (0.715), trucks (0.826), and buses (0.804), achieving an overall $\text{mAP}@50 = 0.519$.

The observed performance gain results from the integration of the Efficient Hybrid Vision Transformer (EHViT) for enhanced feature extraction, the Cross-Dimensional Multi-Scale Attention (CDMSA) and Contextual Multi-Attention Fusion (CMAF) modules for adaptive contextual reasoning, and the Multi-Head Dynamic Attention (MHDA) head for improved localization precision. Collectively, these components enable YOLOv8s-WAMNet to achieve robust and efficient vehicle detection under diverse weather conditions while maintaining real-time performance.

4.5.1 Qualitative Analysis of Detection Performance on WEDGE dataset

To further illustrate the effectiveness of YOLOv8s-WAMNet, Fig. 6 presents a visual comparison of detection results between the baseline YOLOv8s model and the proposed YOLOv8s-WAMNet. The images are selected from the WEDGE dataset test set, representing

challenging real-world scenarios with complex backgrounds. Fig. 6a and 6b show two sample input images. Fig. 6c and 6e display the detection results obtained using YOLOv8s, while Fig. 6d and 6f present the corresponding results from YOLOv8s-WAMNet. The results indicate that YOLOv8s-WAMNet effectively addresses the limitations of YOLOv8s, particularly in detecting occluded and distant objects.



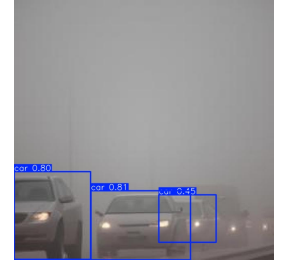
(a) Input WEDGE image 1



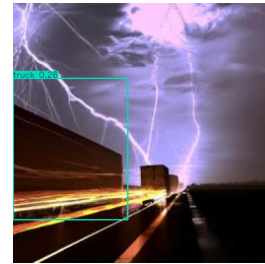
(b) Input WEDGE image 2



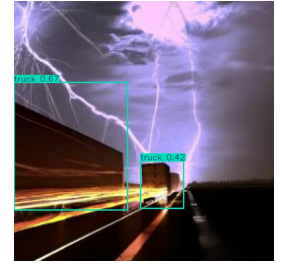
(c) YOLOv8s results



(d) YOLOv8s-WAMNet results



(e) YOLOv8s results



(f) YOLOv8s-WAMNet results

Fig. 6: Qualitative comparison of YOLOv8s and YOLOv8s-WAMNet on WEDGE test images under adverse weather.

Table 8: Comparison of detectors on the WEDGE dataset. Classical baselines report mAP; modern detectors report mAP@50 and mAP@50–95.

Model	mAP (%)	mAP@50 (%)	mAP@50–95 (%)
Faster R-CNN (ResNet-50)	45.41	—	—
MobileNet-Large	43.30	—	—
MobileNet-Large-320	34.98	—	—
YOLOv8s	—	45.60	22.60
YOLOv9s	—	38.30	17.70
YOLOv10s	—	38.80	18.70
YOLOv11s	—	45.70	23.20
YOLOv12s	—	44.60	22.12
DETR	—	48.50	23.10
Deformable DETR	—	50.20	24.20
Conditional DETR	—	49.70	23.80
DN-DETR	—	50.80	24.30
YOLOv8s-WAMNet (Proposed)	—	51.90	24.96

Table 9: Class-wise mAP@50 comparison of YOLOv8s-WAMNet with benchmark models on the WEDGE dataset.

Model	Car	Truck	Bus	Motorcycle	Bicycle	Person	Van	mAP@50
YOLOv8s	0.612	0.760	0.743	0.343	0.000	0.630	0.100	0.456
YOLOv9s	0.506	0.644	0.602	0.354	0.000	0.547	0.030	0.383
YOLOv10s	0.536	0.687	0.676	0.154	0.000	0.551	0.109	0.388
YOLOv11s	0.582	0.747	0.722	0.445	0.000	0.622	0.079	0.457
YOLOv12s	0.593	0.732	0.738	0.211	0.000	0.605	0.244	0.446
DETR	0.662	0.772	0.751	0.428	0.000	0.642	0.139	0.485
Deformable DETR	0.701	0.788	0.765	0.451	0.000	0.660	0.152	0.502
Conditional DETR	0.694	0.781	0.758	0.443	0.000	0.655	0.147	0.497
DN-DETR	0.708	0.796	0.772	0.460	0.000	0.667	0.153	0.508
YOLOv8s-WAMNet (Proposed)	0.715	0.826	0.804	0.462	0.000	0.738	0.197	0.519

4.5.2 Quantitative Analysis on the WEDGE Dataset

Fig. 7 presents a visual quantitative comparison between the standard YOLOv8s and the proposed YOLOv8s-WAMNet on the WEDGE dataset using F1–Confidence curves, Precision–Recall (PR) curves, and normalized confusion matrices.

From the F1–Confidence curves, the baseline YOLOv8s achieves a peak F1 score of approximately 0.38 at a confidence threshold around 0.36. This behavior reflects a relatively conservative detection tendency, where higher confidence thresholds are required to balance precision and recall, particularly under adverse weather conditions.

In contrast, YOLOv8s-WAMNet attains a higher peak F1 score of 0.47 at a substantially lower confidence threshold of 0.138. This shift indicates that the proposed model maintains a more favorable precision–recall balance at lower confidence levels, enabling more stable and reliable detections in challenging multi-weather scenarios.

The Precision–Recall curves further emphasize this improvement. The baseline YOLOv8s achieves

an overall mAP@50 of 0.456, whereas YOLOv8s-WAMNet improves this value to 0.519. Notable gains are observed for vehicle-related classes, including truck (0.760 to 0.820), bus (0.743 to 0.803), and person (0.630 to 0.738), demonstrating enhanced detection robustness and class separability.

The normalized confusion matrices reveal that YOLOv8s exhibits moderate inter-class confusion, particularly between vehicles and background regions. By comparison, YOLOv8s-WAMNet shows stronger diagonal dominance for key classes such as car (0.64), bus (0.74), and truck (0.80), along with reduced confusion with the background class. This improvement confirms more accurate inter-class discrimination and effective suppression of false positives.

Overall, the combined analysis of F1–Confidence behavior, Precision–Recall characteristics, and confusion patterns demonstrates that YOLOv8s-WAMNet consistently outperforms the baseline YOLOv8s. These gains can be attributed to the integration of the Efficient Hybrid Vision Transformer (EHViT), Cross-Dimensional Multi-Scale Attention (CDMSA), Contextual Multi-Attention Fusion (CMAF), and the Multi-Head Dynamic Attention (MHDA) head, which

collectively enhance feature representation, contextual reasoning, and localization accuracy under adverse weather conditions.

4.6 Performance Evaluation on the DAWN Dataset

This subsection evaluates the real-world generalization capability of the proposed YOLOv8s-WAMNet using the DAWN dataset, which contains naturally captured traffic scenes under severe adverse weather conditions. Both quantitative metrics and qualitative visualizations are analyzed to assess detection robustness and practical deployment reliability beyond synthetic benchmarks.

4.6.1 Quantitative Analysis on the DAWN Dataset

Table 10 reports the quantitative performance of the proposed YOLOv8s-WAMNet on the DAWN dataset, which consists of real-world traffic scenes captured under naturally occurring adverse weather conditions. Compared with the YOLOv8s baseline, YOLOv8s-WAMNet improves mAP@50 from 42.7% to 51.0%, yielding a gain of 8.3 percentage points. In addition, the mAP@50–95 metric increases from 27.8% to 30.5%, indicating improved localization accuracy under stricter IoU thresholds.

The quantitative trends are further illustrated in Fig. 9. Specifically, the F1–confidence curves in Fig. 9(a,d) show that YOLOv8s-WAMNet attains a higher optimal F1 score and maintains greater stability across confidence thresholds compared to the baseline. The Precision–Recall curves in Fig. 9(b,e) demonstrate that the proposed model preserves higher precision over a wider recall range, reflecting improved detection reliability in adverse weather conditions. Furthermore, the normalized confusion matrices in Fig. 9(c,f) reveal enhanced class-wise prediction consistency and reduced misclassification among visually similar categories.

These quantitative results confirm that the proposed weather-adaptive architecture generalizes effectively from synthetic training data to real-world adverse environments.

Table 10: Quantitative performance comparison on the DAWN real-world adverse weather dataset.

Model	Prec.	Rec.	mAP@50 / 50–95
YOLOv8s (Baseline)	0.48	0.34	42.7 / 27.8
YOLOv8s-WAMNet (Proposed)	0.55	0.41	51.0 / 30.5

4.6.2 Qualitative Analysis Under Extreme Adverse Weather on the DAWN Dataset

Figure 8 illustrates representative qualitative detection results on the DAWN dataset under three challenging real-world adverse weather conditions: heavy rain, snowstorm, and dense fog. For each scenario, the first row shows the original DAWN input image, the second row presents predictions generated by the YOLOv8s baseline, and the third row reports results obtained using the proposed YOLOv8s-WAMNet.

Visual inspection indicates that YOLOv8s-WAMNet produces more complete and coherent detections compared to the baseline, particularly for distant, partially occluded, and low-contrast vehicles. In rain and fog scenarios, where visibility degradation and background clutter are severe, the proposed model exhibits improved object continuity and more reliable bounding box placement.

Despite these improvements, both models continue to face challenges under extreme conditions, such as near-zero visibility and strong motion blur, where small or heavily occluded vehicles may remain undetected. These qualitative observations highlight both the strengths of the proposed architecture in adverse environments and its current limitations, providing insight into potential directions for future enhancement.

4.7 Real-Time Detection Setup and Performance Evaluation

A real-time object detection system was developed for the proposed YOLOv8s-WAMNet model using Visual Studio Code (VS Code) and Python 3.12.2. The implementation leverages the OpenCV library for video processing and the Ultralytics YOLO framework for loading and deploying optimized model weights. The workflow was designed to achieve efficient real-time inference while maintaining system stability and throughput.

The detection pipeline begins by loading the optimized YOLOv8s-WAMNet weight file (e.g., `last.pt`), which contains fine-tuned parameters

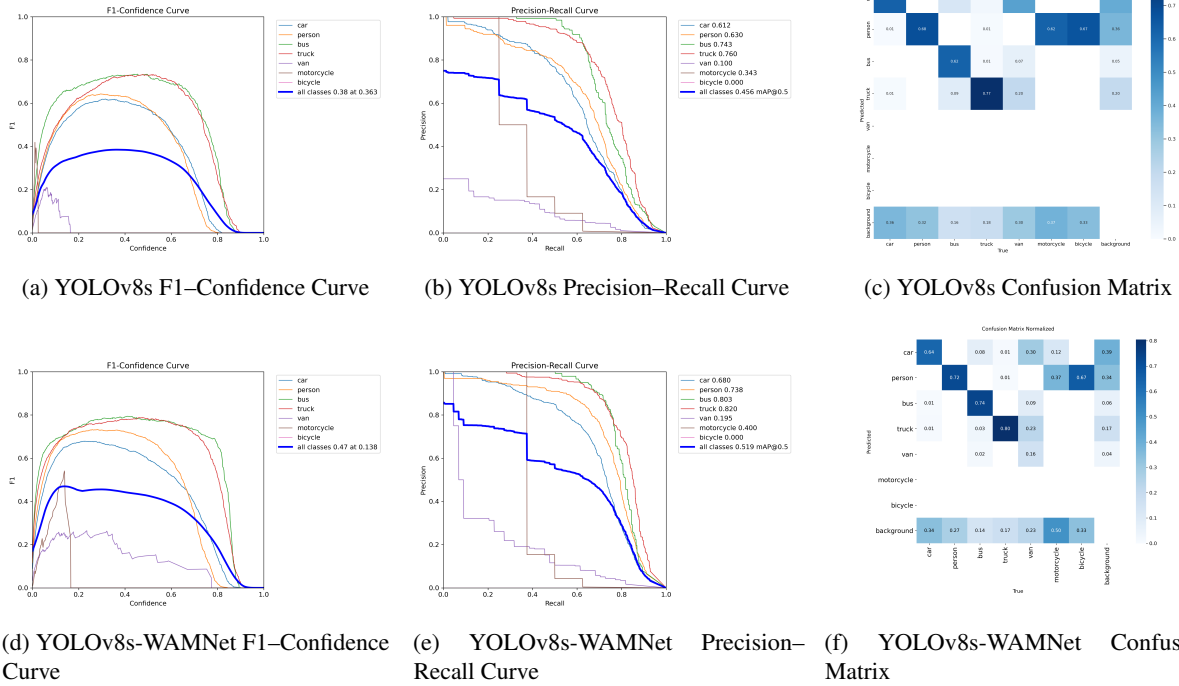


Fig. 7: Visual quantitative comparison between the baseline YOLOv8s and the proposed YOLOv8s-WAMNet on the WEDGE dataset, including F1-Confidence curves, Precision-Recall curves, and normalized confusion matrices.

obtained from extensive training on the WEDGE aerial-vehicle dataset. The model is initialized through the `ultralytics.YOLO` module, ensuring seamless integration within the Python environment.

To enable real-time inference, video frames are captured from a live webcam feed or pre-recorded video file using the `cv2.VideoCapture` interface. Each input stream is processed frame by frame, with both raw and annotated frames saved for post-analysis. Two output directories are created: one for storing unprocessed frames and another for saving detection results with bounding-box overlays.

Following video initialization, key frame properties such as width, height, and frame rate are retrieved and utilized to configure video writers via the `cv2.VideoWriter` function. This configuration ensures synchronized saving of original and processed frames into AVI-format video files.

The detection loop continues until the end of the video stream or manual termination by the user. Each captured frame is first stored in its raw form, then passed to the YOLOv8s-WAMNet model for inference. The model predicts object locations and confidence scores, overlays bounding boxes on detected instances,

and saves the annotated outputs. For visualization, each processed frame is concatenated with its original counterpart using `cv2.hconcat`, providing a side-by-side real-time comparison. The user may terminate the process by pressing the `q` key. Upon completion, all video capture and writer resources are released to prevent memory leaks. The resulting frame sequences and annotated video files facilitate performance evaluation and detailed analysis. This implementation establishes an efficient and structured pipeline for deploying YOLOv8s-WAMNet in real-time applications.

The real-time performance of YOLOv8s-WAMNet was evaluated under two distinct illumination scenarios—summer daylight and night-time—to assess its robustness to varying environmental conditions.

Under summer daylight conditions, characterized by bright sunlight, high ambient temperatures, and intermittent shadows that can distort feature extraction, the model demonstrated stable and reliable detection performance. As shown in Figure 10, each column depicts an original frame (top row) alongside its detection output (bottom row). Specifically, subfigures 10a, 10b, and 10c illustrate representative real-time vehicle and pedestrian detection results in dense urban settings.



Fig. 8: Qualitative comparison on the DAWN dataset under extreme adverse weather conditions. From top to bottom: DAWN input images, YOLOv8s baseline predictions, and YOLOv8s-WAMNet predictions. Results are shown for heavy rain, snowstorm, and fog scenarios. The proposed YOLOv8s-WAMNet achieves more complete detections and improved confidence stability compared to the baseline under severe visibility degradation.

The precise bounding boxes highlight the model's ability to maintain localization accuracy despite lighting variations. The proposed YOLOv8s-WAMNet achieved an average inference rate of approximately

105 FPS (equivalent to about 9.5 ms per frame), confirming its suitability for real-time aerial surveillance and traffic monitoring applications.

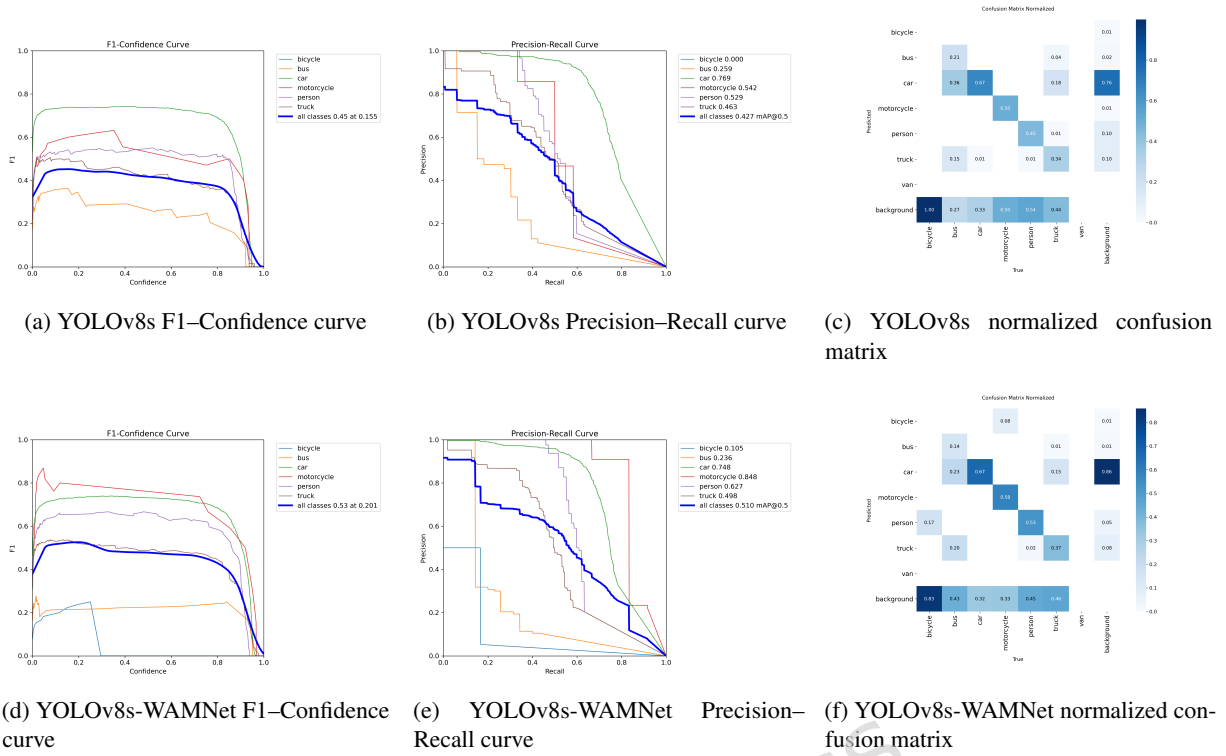


Fig. 9: Quantitative performance visualization on the DAWN dataset. The first row corresponds to the baseline YOLOv8s model, while the second row presents the proposed YOLOv8s-WAMNet. From left to right, F1–confidence curves, precision–recall curves, and normalized confusion matrices are shown, highlighting the improved detection robustness of YOLOv8s-WAMNet under real-world adverse weather conditions.

In contrast, under night-time conditions—characterized by low illumination, non-uniform artificial lighting, and reduced visual contrast—the model continued to deliver robust performance. As illustrated in Figure 11, each column presents an original night-vision frame (top row) and the corresponding detection output (bottom row). Subfigures 11a, 11b, and 11c display typical results captured under low-light environments. Despite the challenging conditions, YOLOv8s-WAMNet maintained consistent detection accuracy and reliable localization. These findings confirm the model’s robustness under low-visibility scenarios and its capability for continuous vehicle detection and monitoring across both day and night conditions, outperforming earlier YOLO variants in both stability and precision.

5 Conclusion

This study presented YOLOv8s-WAMNet, a lightweight and robust detector tailored for vehicle

detection under adverse weather. The architecture integrates attention-driven components—EHViT, CDMSA, CMAF, and MHDA—to enhance multi-scale feature representation and improve detection reliability across challenging illumination and visibility conditions.

Experimental results on the WEDGE dataset demonstrate that YOLOv8s-WAMNet achieves superior detection performance with $\text{mAP}@50 = 51.90\%$ and $\text{mAP}@50-95 = 24.96\%$, surpassing both classical and modern detectors such as YOLOv8s, YOLOv11s, and Deformable DETR. With only 9.69 M parameters and 13.24 GFLOPs, the model maintains an excellent balance between accuracy and computational efficiency, achieving consistent detection across diverse weather conditions, including fog, rain, and snow. Additional experiments on the real-world DAWN dataset further confirm the robustness and generalization capability of YOLOv8s-WAMNet under naturally occurring adverse weather conditions. Future work will aim to enhance robustness under extreme weather scenarios and leverage multi-sensor modalities such as thermal



Fig. 10: Real-time detection results of YOLOv8s-WAMNet under summer day conditions. Top row in each column: original frame; bottom: corresponding detection result.



Fig. 11: Real-time detection results under night vision conditions. Each column: original (top) and its detection (bottom).

and radar imagery for improved adaptability. Further optimization through pruning, quantization, and TensorRT acceleration will facilitate deployment on embedded and UAV-based platforms. Additionally, continual learning and domain adaptation strategies will be explored to ensure sustained performance in dynamic real-world environments.

Limitations

Despite the strong performance of YOLOv8s-WAMNet on both the synthetic WEDGE dataset and the real-world DAWN dataset, certain challenges remain. The model exhibits reduced accuracy for rarely occurring and small-scale object categories, such as bicycles, which is mainly attributed to dataset imbalance and limited visual cues under adverse weather conditions.

Although evaluation on the DAWN dataset demonstrates improved real-world generalization, performance degradation is still observed in extremely challenging scenarios involving near-zero visibility, severe motion blur, or heavy occlusions. These conditions continue to pose difficulties for single-frame, vision-only detection frameworks.

The current implementation relies exclusively on RGB image inputs. Incorporating additional sensing modalities, such as thermal, infrared, or radar data, represents a promising direction for further improving robustness in extreme weather and nighttime conditions. Finally, while the proposed architecture is designed to be computationally efficient and suitable for real-time deployment, a comprehensive evaluation on ultra-low-power embedded or edge-specific hardware platforms remains an avenue for future investigation.

Statements and Declarations

Funding

The authors did not receive support from any organization for the submitted work. No funding was received to assist with the preparation of this manuscript.

Competing Interests

The authors declare that they have no relevant financial or non-financial interests to disclose.

Author Contributions

All authors contributed to the conception and design of the study. Manjit Jaiswal carried out the implementation, experiments, and drafting of the manuscript. Kapil Kumar Nagwanshi supervised the work and reviewed the methodology. Maleika Heenaye-Mamode Khan contributed to analysis and manuscript revision. Upen-dra Verma assisted in data preparation, model training, and visualization. Amelia Taylor contributed to result interpretation, language editing, and correspondence. All authors read and approved the final manuscript.

Data Availability

The datasets used in this study are publicly available. The WEDGE dataset, introduced by Marathe, A., Ramanan, D., Walambe, R., and Kotecha, K. (2023) in “WEDGE: A multi-weather autonomous driving dataset built from generative vision-language models,” is available at <https://arxiv.org/abs/2305.07528>.

The DAWN (Detection in Adverse Weather Nature) dataset, proposed by Kenk, M. A. and Hassaballah, M. (2020) for vehicle detection under real-world adverse weather conditions, can be accessed at <https://data.mendeley.com/datasets/766ygrbt8y/3>.

All other data supporting the findings of this study are included within the article. Model weights, source code, and additional experimental materials will be made available upon reasonable request.

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